# pmsimstats Codebase Overview

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## Contents

**1 Core R Package Files (R/)** **1**
1.1 generateData.R . . . 1
1.2 generateSimulatedResults.R . . . 2
1.3 lme_analysis.R . . . 2
1.4 carryover_analysis.R . . . 2
1.5 buildtrialdesign.R . . . 3
1.6 utilities.R . . . 3
1.7 censordata.R . . . 3
1.8 plottingfunctions.R . . . 3

**2 Analysis Scripts** **3**
2.1 Figure 4 (analysis/figure4/) . . . 3
2.2 Figure 5 (analysis/figure5/) . . . 3
2.3 Extracted Code by Commit . . . 3

**3 White Papers (docs/pdf/)** **4**

**4 Bundled Data (data/)** **4**

**5 Key Parameters** **5**
5.1 Current Configuration (Revised Code) . . . 5
5.2 Publication Configuration (Original Code) . . . 5

**6 Dependencies** **5**

The `pmsimstats` R package implements Monte Carlo simulation-based power analysis for aggregated N-of-1 clinical trial designs, comparing four trial designs on their ability to detect a biomarker-treatment interaction. Based on Hendrickson et al. (2020), *Frontiers in Digital Health* 2:13.

# 1 Core R Package Files (R/)

## 1.1 generateData.R

The data-generating process (DGP). Two exported functions:

- **`generateData()`** – Draws simulated participant data from a multivariate normal distribution. Accepts a pre-built covariance matrix (`cached_sigma`) or builds one via `buildSigma()`. Uses

pre-computed Cholesky factor for efficient MVN draws (`L %*% Z + mu`). Computes outcome columns via vectorized `data.table` operations.

Under Architecture A (`dgp_architecture = 'mean_moderation'`), the BM-BR correlation block is omitted from $\Sigma$ and the biomarker-treatment interaction is applied as an additive post-draw step: $\text{BR}_{it} \mathrel{+}= c_{bm} \cdot B_i^* \cdot \mathbf{1}[\text{on drug}] \cdot \sigma_{\text{BR}}$, where $B_i^*$ is the standardized biomarker.

- **`buildSigma()`** – Constructs the 26x26 covariance matrix for a single trial path. Implements:
  - AR(1) within-factor autocorrelation: $\rho^{|t_i - t_j|}$
  - Cross-factor correlations (same-time and different-time with AR(1) decay)
  - Biomarker-BR correlation with exponential decay during off-drug periods: $c.bm \cdot \exp(-\lambda_{cor} \cdot tsd)$
  - Auto-derives $\lambda_{cor} = \ln(2)/\text{carryover_t1half}$
  - Positive definiteness check with verbose warning
  - Pre-computes Cholesky factor for caching
- **`validateParameterGrid()`** – Tests all parameter combinations for PD before simulation. Reports failures upfront.

## 1.2 generateSimulatedResults.R

The simulation loop. Single exported function:

- **`generateSimulatedResults()`** – Orchestrates the full parameter sweep:
  1. PD pre-validation of all parameter combinations
  2. Sigma matrix caching (builds all unique matrices once)
  3. Parameter grid expansion across designs, response params, baseline params, and model params
  4. Per-parameter-set worker function (`run_one_paramset`) that generates data, fits LME, applies censoring, and collects results
  5. Parallel execution via `furrr::future_map()` with closure-based function passing to workers
  6. Progressive save support for long runs

  Key parameters: `lambda_cor` (correlation decay, default NA = auto), `n_cores` (parallelism, default auto-detect).

## 1.3 lme_analysis.R

The analysis model. Single exported function:

- **`lme_analysis()`** – Fits a linear mixed-effects model to simulated or actual trial data:
  - Uses `nlme::lme` with `corCAR1(form = ~t|ptID)` for continuous-time AR(1) residual correlation

  - Dynamic formula selection: `Sx ~ bm + t + Dbc + bm:Dbc` when within-subject drug variation exists; `Sx ~ bm + t + bm:t` for open-label
  - `Dbc` is a continuous drug indicator (1 when on drug, exponential decay when off drug)
  - Graceful handling of rank-deficient models (returns NA)
  - Fallback to `lme` without `corCAR1` if optimization fails
  - Extracts interaction coefficient from `nlme` tTable

### 1.4 carryover_analysis.R

Extended analysis functions for actual trial data:

- **`characterize_carryover()`** – Sweeps candidate carryover half-lives (default 0.1 to 8 weeks), fits `nlme::lme` at each, compares AIC/BIC to identify the best-fitting half-life. Returns drug effect, interaction, AR(1) phi at each half-life.
- **`analyze_trial_extended()`** – Extracts both the main drug effect (`Dbc`) and biomarker interaction (`bm:Dbc`) with 95% CIs. Computes variance components, predicted drug effect as a function of biomarker value, and clinical decision threshold.
- **`print_carryover_summary()`** / **`print_trial_summary()`** – Clinical reporting summaries.

### 1.5 buildtrialdesign.R

Trial design specification:

- **`buildtrialdesign()`** – Constructs trial paths from intuitive inputs (timepoints, expectancies, on-drug vectors). Computes derived timing variables: `tod` (time on drug), `tsd` (time since discontinuation), `tpb` (time in positive belief).

### 1.6 utilities.R

Helper functions:

- **`modgompertz()`** – Modified Gompertz function for response trajectories. Passes through origin, asymptotes at `maxr`.
- **`cumulative()`** – Converts interval durations to cumulative time.
- **`reknitsimresults()`** – Recombines progressive save chunks.

### 1.7 censordata.R

Dropout simulation:

- **`censordata()`** – Two-component dropout: time-dependent (flat rate) and outcome-dependent (biased by symptom change). Implements last-observation-carried-forward censoring.

### 1.8 plottingfunctions.R

Visualization:

- **`PlotModelingResults()`** – Power heatmaps across 4-dimensional parameter grids with `ggplot2` `geom_tile` and `facet_grid`.
- **`plotfactortrajectories()`** – Factor trajectory plots by trial path.

# 2 Analysis Scripts

## 2.1 Figure 4 (`analysis/figure4/`)

| Script | Purpose |
|---|---|
| `01-generate-power-sweep.R` | Figure 4: current code, sequential |
| `02-generate-power-sweep-original.R` | Figure 4: original commit code |
| `03-generate-power-by-commit.R` | Figure 4: parameterized by commit |
| `04-generate-power-head.R` | Figure 4: current code, parallel, c.bm={0,0.25,0.45} |
| `05-pd-diagnostics.R` | PD failure rate instrumentation |
| `06-plot-power-heatmaps.R` | Heatmap plotting with provenance |

## 2.2 Figure 5 (`analysis/figure5/`)

| Script | Purpose |
|---|---|
| `01-generate-parameter-sensitivity.R` | Figure 5: revised code |
| `02-generate-parameter-sensitivity-original.R` | Figure 5: original code |

## 2.3 Extracted Code by Commit

Code from each historical commit extracted for comparison:

```
analysis/scripts/commit_42ac030/   # Initial (publication)
analysis/scripts/commit_8609f12/   # Ron Thomas edits
analysis/scripts/commit_f6ee86d/   # Autocorrelation typo fix
analysis/scripts/commit_f70f86d/   # Carryover added
analysis/scripts/commit_325314f/   # Modified Db (broken)
analysis/scripts/original_R/      # Copy of 42ac030
```

# 3 White Papers (`docs/pdf/`)

# 4 Bundled Data (`data/`)

# 5 Key Parameters

## 5.1 Current Configuration (Revised Code)

## 5.2 Publication Configuration (Original Code)

# 6 Dependencies

Core: `data.table`, `nlme`, `lme4`, `lmerTest`, `corpcor`, `MASS`, `ggplot2`

| # | Document | Content |
|---|---|---|
| 01 | Codebase Overview | Package structure, parameters, dependencies |
| 02 | DGP Architecture Comparison | Mean moderation vs. MVN correlation |
| 03 | Audit and Revision Report | Seven DGP corrections, PD resolution, deliverables |
| 04 | Revised Power Analysis | Problems, fixes, validated results |
| 06 | AR(1) Residual Correlation | lmer to nlme transition with Type I error validation |
| 07 | Positive Definiteness Failures | PD failure prevalence and causes |
| 08 | BM-BR Correlation Decay | Pharmacokinetic derivation of $\lambda_{\text{cor}}$ |
| 09 | Carryover Correlation Artifact | BM-BR correlation artifact analysis |
| 10 | Carryover Analysis Assessment | Carryover characterization in actual data |
| 13 | Figure 4 Walkthrough | Pre-audit methodology reference |
| 14 | Trial Analysis Guide | Collaborator goal implementation |
| 15 | Tidyverse Alignment Plan | Alignment between orig and 2025 pipelines |
| 16 | Tidyverse Code Review | Code review of 2025 pipeline |
| 17 | Simplification Plan | Codebase simplification strategy |
| 18 | Response Parameter Sensitivity | Response parameter sensitivity before/after |
| 19 | Mean Moderation Implementation | Implementation derivation for Architecture A |

| File | Content |
|---|---|
| `results_core.rda` | Published Figure 4 results (initial commit, 500 reps) |
| `extracted_bp.rda` | Baseline parameters (BP mean/SD, CAPS mean/SD) |
| `extracted_rp.rda` | Response parameters (Gompertz max/disp/rate/sd) |
| `CTdata.rda` | Actual clinical trial data |
| `results_maxes.rda` | Figure 5A results (response maxima) |
| `results_rates.rda` | Figure 5B results (response rates) |
| `results_trajectories.rda` | Factor trajectory data |

Performance: `furrr`, `future` (parallel processing)

Legacy (in DESCRIPTION but used minimally): `svMisc`, `tictoc`, `ggpubr`, `gridExtra`

| Parameter | Value | Rationale |
|---|---|---|
| Autocorrelation | AR(1), rho = 0.7 | PD-feasible, clinically realistic |
| c.bm | {0, 0.25, 0.45} | Max PD-feasible = 0.45 (OL+BDC) |
| | | Under Architecture A, the PD constraint on $c_{bm}$ is removed |
| Carryover t_half | {0, 0.5, 1.0} weeks | Clinically realistic |
| lambda_cor | ln(2) / t_half (auto) | Pharmacokinetic derivation |
| Cross-factor (same time) | c.cf1t = 0.2 | Unchanged from publication |
| Cross-factor (diff time) | c.cfct * $\rho^{\lvert t_i - t_j \rvert}$ | AR(1) decay applied |
| Analysis model | nlme::lme + corCAR1 | Required for AR(1) Type I error control |
| PD corrections | 0/162 (0%) | All matrices valid |

| Parameter | Value | Issue |
|---|---|---|
| Autocorrelation | CS, rho = 0.8 | PD failures at c.bm > 0.25 |
| c.bm | {0, 0.3, 0.6} | 0.6 exceeds PD-feasible range |
| Carryover t_half | {0, 0.1, 0.2} weeks | Too short for genuine effects |
| BM-BR correlation | Step function | Power drop artifact |
| Analysis model | lmer (no corCAR1) | Correct under CS |
| PD corrections | 40/162 (24.7%) | Silent, uncontrolled |